SUMMATIVE ASSESSMENT – 1 (SA-1) MODEL PAPER – SET 3

CLASS IX · MATHEMATICS · Semester-1 · AP SCERT 2026-27 · SSC 100-Mark Pattern

BLUEPRINT – SA-1 MODEL PAPER – SET 3

SUBJECT: MATHEMATICS	CLASS: IX	TIME: 3 Hrs 15 Min	MAX. MARKS: 100

Syllabus: Class IX Mathematics, Semester-1 (AP SCERT 2026-27) – 1. Number Systems, 2. Polynomials, 3. Coordinate Geometry, 4. Linear Equations in Two Variables, 5. Introduction to Euclid's Geometry, 6. Lines and Angles. Pattern: SSC 100-mark Summative Assessment blueprint with Academic Standards (Awareness 60%, Sensitivity 20%, Creativity 20%).

1. Weightage to Academic Standards (Ability)

ABILITY	AWARENESS (60%)	SENSITIVITY (20%)	CREATIVITY (20%)	TOTAL
MARKS	60	20	20	100

2. Weightage to Form of Questions

FORM OF QUESTION	Essay / Long Ans.	Short Answer	Very Short Ans.	Objective	TOTAL
No. of questions	5	8	8	12	33
Marks allotted	8	4	2	1	100
Estimated time	70 min	60 min	30 min	20 min	180 + 15 min

3. Ability-wise Distribution of Questions

ABILITY	1 Mark	2 Marks	4 Marks	8 Marks	Total Marks
Awareness	4	4	6	3	60
Sensitivity	4	2	1	1	20
Creativity	4	2	1	1	20
Total Questions	12	8	8	5	100

4. Chapter-wise Weightage

S.No	UNIT / CHAPTER	MARKS
1	Number Systems	1+1+2+2+4+4+8+(8) = 22 + (8)
2	Polynomials	1+1+2+4+4+8+(8) = 20 + (8)
3	Linear Equations in Two Variables	1+1+2+4+8+(8) = 16 + (8)
4	Coordinate Geometry	1+1+2+4+8 = 16
5	Introduction to Euclid's Geometry	1+1+2+4+(8) = 8 + (8)
6	Lines and Angles	1+1+2+2+4+8+(8) = 18 + (8)
	TOTAL	**100 + (40)**

Note: Marks within ( ) are the internal-choice (OR) questions of Section IV. The 100 marks are distributed over the six Semester-1 chapters; the ability-wise (60 : 20 : 20) and form-wise weightage follow the SSC blueprint pattern.

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5. Question-wise Chapter Distribution and Domain (Questions 1 to 33)

Q.No	Chapter / Topic	Domain	Marks
1	Number Systems	Awareness	1
2	Polynomials	Creativity	1
3	Linear Equations in Two Variables	Creativity	1
4	Coordinate Geometry	Creativity	1
5	Introduction to Euclid’s Geometry	Sensitivity	1
6	Lines and Angles	Awareness	1
7	Number Systems	Sensitivity	1
8	Polynomials	Sensitivity	1
9	Introduction to Euclid’s Geometry	Creativity	1
10	Coordinate Geometry	Awareness	1
11	Linear Equations in Two Variables	Awareness	1
12	Lines and Angles	Sensitivity	1
13	Number Systems	Creativity	2
14	Number Systems	Awareness	2
15	Polynomials	Awareness	2
16	Coordinate Geometry	Sensitivity	2
17	Linear Equations in Two Variables	Awareness	2
18	Introduction to Euclid’s Geometry	Sensitivity	2
19	Lines and Angles	Awareness	2
20	Lines and Angles	Creativity	2
21	Number Systems	Awareness	4
22	Number Systems	Awareness	4
23	Polynomials	Awareness	4
24	Polynomials	Awareness	4
25	Linear Equations in Two Variables	Sensitivity	4
26	Coordinate Geometry	Awareness	4
27	Introduction to Euclid’s Geometry	Awareness	4
28	Lines and Angles	Creativity	4
29a / 29b	(a) Number Systems (b) Introduction to Euclid’s Geometry	Creativity	8
30a / 30b	(a) Polynomials (b) Number Systems	Awareness	8
31a / 31b	(a) Linear Equations in Two Variables (b) Polynomials	Sensitivity	8
32a / 32b	(a) Lines and Angles (b) Linear Equations in Two Variables	Awareness	8
33a / 33b	(a) Coordinate Geometry (b) Lines and Angles	Awareness	8

ABILITIES AND THEIR OBJECTIVES AS PER PARAKH NORMS

I) AWARENESS: Knowledge / understanding of the construct / understanding of the process (resembles old objectives Knowledge, Understanding and Application).

II) SENSITIVITY: Conflict resolution / open-mindedness / collaboration (resembles old objective Analysis).

III) CREATIVITY: Generate innovative ideas / flexibility / fluency / exploration / combining ideas and concepts (resembles old objectives Evaluation and Creativity).

NOTE: Overall, the chapter-wise and ability-wise weightage is followed. There is no rigid rule that a question with a particular number must come from a particular chapter.

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SUMMATIVE ASSESSMENT – 1 : MODEL QUESTION PAPER – SET 3

MATHEMATICS – CLASS IX (Semester-1, AP SCERT Syllabus 2026-27)

Time : 3 Hrs 15 Min (15 min reading + 3 hrs writing) **Max. Marks : 100**

General Instructions: *(i) Answer all the questions. (ii) Write the answers neatly in the answer booklet. (iii) The paper has four Sections (I, II, III, IV). (iv) Use of calculators is not permitted. (v) Draw neat, labelled figures wherever necessary.*

SECTION – I 12 × 1 = 12 Marks

Note : *i) Answer all the questions. ii) Each question carries 1 mark. iii) Choose the correct option and write its letter (A/B/C/D) in the answer booklet.*

1. The decimal expansion of √2 is *[Awareness · Number Systems]*

(A) terminating (B) non-terminating recurring
(C) non-terminating non-recurring (D) none of these

2. If (x + 2) is a factor of $x^3 + 3x^2 + kx + 6$, then k = *[Creativity · Polynomials]*

(A) 3 (B) 4 (C) 5 (D) 6

3. The equation of the line parallel to the x-axis passing through (2, −3) is *[Creativity · Linear Equations in Two Variables]*

(A) x = 2 (B) y = −3 (C) x = −3 (D) y = 2

4. The point which is 3 units to the right of the y-axis and 2 units below the x-axis is *[Creativity · Coordinate Geometry]*

(A) (3, −2) (B) (−3, 2) (C) (2, −3) (D) (−2, 3)

5. Ravi says 'All right angles are equal to one another' is a postulate, while Sita says it is a theorem. Who is correct? *[Sensitivity · Introduction to Euclid's Geometry]*

(A) Ravi (B) Sita (C) Both (D) None

6. Two complementary angles are such that one is twice the other. The angles are *[Awareness · Lines and Angles]*

(A) 45°, 45° (B) 30°, 60° (C) 20°, 70° (D) 40°, 50°

7. The simplest form of 1/(√3 + √2) is *[Sensitivity · Number Systems]*

(A) √3 + √2 (B) √3 − √2 (C) 1/5 (D) 5

8. The coefficient of x^2 in the expansion of $(x + 2)^3$ is *[Sensitivity · Polynomials]*

(A) 2 (B) 4 (C) 6 (D) 8

9. A, B, C lie on a line with B between A and C. If AB = 4 cm and BC = 5 cm, then AC = *[Creativity · Introduction to Euclid's Geometry]*

(A) 1 cm (B) 9 cm (C) 20 cm (D) 5 cm

10. The point (0, −4) lies on the *[Awareness · Coordinate Geometry]*

(A) positive x-axis (B) negative x-axis (C) positive y-axis (D) negative y-axis

11. Which of the following is a linear equation in two variables? *[Awareness · Linear Equations in Two Variables]*

(A) $x^2 + y = 3$ (B) 2x + 3y = 7 (C) xy = 4 (D) x + 1/y = 2

12. Two parallel lines are cut by a transversal. If one of the interior angles on the same side is 70°, the other is *[Sensitivity · Lines and Angles]*

(A) 70° (B) 110° (C) 20° (D) 290°

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SECTION – II 8 × 2 = 16 Marks

Note : *i) Answer all the questions. ii) Each question carries 2 marks.*

13. Find an irrational number between √2 and √3. *[Creativity · Number Systems]*

14. Simplify: $(\sqrt{5} + \sqrt{7})^2$. *[Awareness · Number Systems]*

15. Find the remainder when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 2)$. *[Awareness · Polynomials]*

16. Priya says every point on the x-axis has ordinate 0, while Raju says it has abscissa 0. Who is correct? *[Sensitivity · Coordinate Geometry]*

17. Which of (2, −1) and (1, 1) is a solution of $3x + 2y = 4$? *[Awareness · Linear Equations in Two Variables]*

18. Rekha says 'A point is that which has no part' is a postulate of Euclid, while Rani says it is a definition. Who is correct? *[Sensitivity · Introduction to Euclid's Geometry]*

19. Two angles of a triangle are 50° and 65°. Find the third angle and name the type of triangle. *[Awareness · Lines and Angles]*

20. ∠AOC and ∠BOC form a linear pair in the ratio 7 : 5. Find ∠AOC. *[Creativity · Lines and Angles]*

SECTION – III 8 × 4 = 32 Marks

Note : *i) Answer all the questions. ii) Each question carries 4 marks.*

21. Rationalise: (i) $1/(\sqrt{5} + 2)$ (ii) $4/(\sqrt{7} - \sqrt{3})$. *[Awareness · Number Systems]*

22. Simplify: (i) $8^{2/3}$ (ii) $81^{-1/4}$ (iii) $2^{1/3} \times 2^{1/6}$ (iv) $2^{1/2} \times 3^{1/2}$. *[Awareness · Number Systems]*

23. Factorise: (i) $6x^2 + 5x - 6$ (ii) $x^3 - 7x + 6$. *[Awareness · Polynomials]*

24. Using suitable identities evaluate: (i) 98 × 102 (ii) 105 × 107. *[Awareness · Polynomials]*

25. The sum of a number x and twice another number y is 10. Write a linear equation. Ravi says (4, 3) is a solution and Sita says (2, 5) is a solution. Who is correct? *[Sensitivity · Linear Equations in Two Variables]*

26. Plot A(−3, 0), B(3, 0) and C(0, 4) on a graph paper, join them and name the figure. Find its area. *[Awareness · Coordinate Geometry]*

27. Write any four of Euclid's definitions. *[Awareness · Introduction to Euclid's Geometry]*

28. A transversal cuts two parallel lines. A pair of corresponding angles are $(3x + 5)°$ and $(5x - 25)°$. Find x and the angles. *[Creativity · Lines and Angles]*

SECTION – IV 5 × 8 = 40 Marks

Note : *i) Answer all the questions. ii) Each question carries 8 marks. iii) There is an internal choice for each question.*

29.

a) i) Find six rational numbers between 1/4 and 1/3.
ii) Represent √3 on the number line (write the steps of construction). *[Creativity · Number Systems]*

OR

b) i) A, B, C, D are four points on a line in that order. If AB = CD, prove that AC = BD.
ii) C is the mid-point of AB and D is the mid-point of AC. Prove that AD = ¼AB. *[Creativity · Introduction to Euclid's Geometry]*

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30.

a) Factorise $x^3 - 3x^2 - 10x + 24$ using the factor theorem and hence write its zeroes. *[Awareness · Polynomials]*

OR

b) If $a = 2 + \sqrt{3}$, find (i) $1/a$ (ii) $a + 1/a$ (iii) $a^2 + 1/a^2$ (iv) $a - 1/a$. *[Awareness · Number Systems]*

31.

a) Sam says the graph of $y = 2x$ passes through the origin but Tina says it does not. Resolve the conflict by preparing a table of values and drawing the graph. From the graph find y when $x = 3$ and x when $y = -4$. *[Sensitivity · Linear Equations in Two Variables]*

OR

b) i) If $x + y + z = 0$, show that $x^3 + y^3 + z^3 = 3xyz$.
ii) Hence evaluate $12^3 + (-5)^3 + (-7)^3$. *[Sensitivity · Polynomials]*

32.

a) i) If two parallel lines are cut by a transversal, prove that each pair of alternate interior angles is equal.
ii) Two parallel lines are cut by a transversal. Two interior angles on the same side are $(3x + 10)°$ and $(2x + 20)°$. Find x and the two angles. *[Awareness · Lines and Angles]*

OR

b) Draw the graphs of $x + y = 4$ and $x - y = 2$ on the same graph paper. Find the solution and the area of the triangle formed by the two lines and the y-axis. *[Awareness · Linear Equations in Two Variables]*

33.

a) Plot P(2, 2), Q(−2, 2), R(−2, −2), S(2, −2) and T(0, 4). Join P to Q, Q to R, R to S, S to P, P to T and T to Q. (i) Name the figure PQRS. (ii) Find its area. (iii) Find the area of ΔPQT. (iv) Find the total area of the figure. *[Awareness · Coordinate Geometry]*

OR

b) i) In ΔABC, the bisectors of ∠B and ∠C meet at O. Prove that $\angle BOC = 90° + \frac{1}{2}\angle A$. Hence find ∠BOC if $\angle A = 70°$.
ii) The angles of a triangle are in the ratio 3 : 4 : 5. Find the angles. *[Awareness · Lines and Angles]*

— ALL THE BEST —

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